Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department

Academic Year	2016-2017
Semester	First
Exam Date	2 /1 / 2017

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject:	(E501) Digital Signal Processing	**Spec.:**	5th Elec. Eng. & Comm. Tech. Dep.
Examiners:	Dr. Samir Kamal and Prof. Mostafa Afifi	**Time:**	3 hours

Number of Pages: 2 Number of Questions: 4 Attempt all questions

Question 1 (25 points)

a - In detail, what are the main characteristics of the discrete time sinusoidal signals?

b- Check the periodicity of the following sequences and find their fundamental periods.

(i) $y(t) = 7\cos(400\pi t)$ (ii) $x(n) = \sin(2n + \pi/4)$

(c) For the signal $x(t) = 2\cos(3000\pi t)$ to be sampled with sampling frequency F_S = 3 KHz.

(i) Is F_S suitable for sampling the given signal, $x(t)$,

(ii) If it is suitable, use it to find an expression for the corresponding sequence, x(n),

(iii) Find the values of x(n) in the first cycle and

(iv) Plot (showing the steps) the given $x(t)$ and the obtained x(n) on the same figure.

Question 2 (20 points)

a- Find the relationship between the frequency of analog signal, frequency of discrete-time signal and sampling frequency.

b- Show that the DFT, X(K), of a sequence x(n) is periodic.

c- For the following sequence: x(n) = {1, 0, 1, 1}:

(i) Using the direct method, DFT, find the frequency components of this sequence.
(ii) Check your results obtained in part (i) by calculating x(n) using the IFFT algorithm.
(iii) Using your results, verify Parseval's theorem.
(iv) Plot the amplitude and the phase spectrums.
(v) What is the computational saving if we use FFT algorithm instead of DFT algorithm.

Question 3 (20 points)

a- Define (using equations and/or figures) the following items: Correlation operation, Correlation coefficient.

b - Given two sets of samples $x_1(n)$ = {1, 2, 1} and $x_2(n)$ = {0, 3, 5, 1}.

(i) Find the linear cross- correlation between $x_1(n)$ and $x_2(n)$.
(ii) Plot the correlation function obtained in part (i) v.s. the lag number, j.
(iii) Find the autocorrelation function of the sequence $x_2(n)$.

c- For the following two sequences x1 (n) and x2 (n), find their convolution, y(n) and plot y(n) v.s n.

$x_1(n)$ = {2, 3, 0, 1} and $x_2(n)$ = {0, 1, 1}

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Question 4 **(25 points)**

a- For LTIS define (using equations and/or figures) the linearity, stability, ROC and difference equation.

b- For the following transfer function, use the partial fraction method to find its impulse response, h(n).

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$$

c- For the transfer function of a digital system given in part (b):

(i) Plot the pole-zero diagram of H(z),

(ii) Discuss the system stability,

(iii) Determine the ROC of this transfer function,

(iv) Obtain the difference equation of this system,

(v) Represent the filtering operation in a block diagram form.

Z-transforms of some required sequences.

Discrete-time sequence x(n), n ≥ 0	*z-transform X(z)*	*Note*
K	$\frac{kz}{z-1}$	k and α are constants
kn	$\frac{kz}{(z-1)^2}$	
kn^2	$\frac{kz(z+1)}{(z-1)^3}$	
$k\alpha^n$	$\frac{kz}{z-\alpha}$	

Good Luck